

Introduction to Machine Learning

What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that allows systems to learn and improve from experience without being explicitly programmed. It uses algorithms to detect patterns in data and make decisions or predictions.

Types of Machine Learning

1. Supervised Learning

- The algorithm is trained on labeled data.
- Example: Predicting house prices based on previous sales.

2. Unsupervised Learning

- The algorithm works on unlabeled data to find patterns.
- Example: Customer segmentation in marketing.

3. Reinforcement Learning

- The algorithm learns by interacting with an environment and receiving feedback.
- Example: Teaching a robot to walk.

Key Algorithms

- Linear Regression
- Decision Trees
- K-Means Clustering
- Neural Networks

Applications

- Spam detection in emails

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- Image recognition
- Medical diagnosis
- Recommender systems (e.g., Netflix)